# A COMPUTERIZED LEXICON OF AMERICAN SIGN LANGUAGE: THE DASL 1965 IN FORTRAN

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**Abstract.**

This paper describes our methods of organizing, storing, and retrieving lexical data from computerized files of American Sign Language. We present the results of sample searches of the dictionary files and discuss how these files and applications might be expanded. Such a computerized lexicon becomes a useful resource for both researchers and teachers of sign languages.

**Objectives.**

We set out to enter the lexicon of American Sign Language (ASL) in computer files in order to develop and maintain an up-to-date data bank, which will contain all relevant information On the vocabulary of this language. This will enable investigations of the phonological and morphological structure of ASL to be made more quickly, thoroughly, and accurately than would otherwise be the case.

There are many reasons to create such a data bank. For example, the frequency with which a given formational unit appears in the lexicon of ASL is sometimes a variable in psycholinguistic experiments (Poizner 1978, Stungis 1980, Grosjean 1979). Likewise, the researcher may want to know how specific locations (LOC), hand configurations (HC), palm and metacarpal orientations (PO, MCO) combine to form signs in the lexicon, and which combinations do not exist. These phonological regularities in ASL can be further probed and analyzed with the computerized dictionary to reveal functional explanations. For example, a given location or a formational type such as two-handedness permits only a subset of the full ASL HC or movement (MOV) repertoire for constructing signs.

**DASL Notation.**

Our project began with A Dictionary of American Sign Language (DASL), compiled by Stokoe, Casterline, and Croneberg (1965). To date it is the largest inventory of ASL, with approximately 2,500 signs listed. It is relatively free also of manual coinages for English words, and it provides genuine English translation, not just the one-word matchings or glosses that are found in nearly all other sign manuals. DASL is convenient for conversion into a computerized code because its entries include a phonemic transcription system developed specifically for ASL by Stokoe (1960).

Stokoe's notation resulted from his analysis of signs into three quasi-simultaneously occurring aspects: handshape, location, movement. The various instances of each aspect are grouped into finite sets of contrasts, resulting in 55 manual "phonemes" he called cheremes. About 50 distinct hand configurations occurring in ASL signs he grouped into 19 HC phonemes. In similar fashion, 13 LOC and 24 MOV phonemes were established; and there is a notation symbol for each of the phonemes.

In addition to the 55 notation symbols there are diacritics to add more "phonetic" detail to the manual contrasts. For example, the LOC phoneme may receive a diacritic meaning 'below the location specified ;' without the diacritical marking the customary position is indicated. The HC and MOV phonemes also can be modified in the Stokoe notation: there are diacritics for thumb extension, finger bending, and a sharp manner of movement, among others. Further diacritics are used to designate the initial position of the two hands relative to each other. Where necessary, MOV symbols are used to specify palm orientation and direction of fingers, as subscripts to HC symbols. Multiple MOV notations are used in DASL for consecutive and simultaneous movements, by arranging the symbols horizontally and vertically, respectively. Notation of simultaneous movements is more economical with a "sum" of two simple movement components than with additional notations for diagonal movements, gliding, grazing, etc. When the forearm is prominently involved in the articulation of a sign the LOC notation for the forearm is attached as a prefix to the HC symbol. Finally, when a sign changes from one HC into another, the final HC produced is shown in brackets after MOV.

**Coding.**

One method of storing signs from DASL in a computer is to match the notational characters one-tone with 255 binary representations of an 8-bit byte (for 64 different DASL characters plus whatever coding is needed to distinguish, for example, a MOV character in first position from the same character in second position), and to place each lexical item on a variable length record of between 3 and 20 characters. A second method is to select a 7-bit ASCII code for each DASL character and distribute the notation of each sign across a fixed length record according to some coding scheme. The second option requires several bytes just to carry space characters, thus "wasting" storage space; however, keeping records of equal length permits the use of random access files with a numbered address for each record. This simplifies writing search and editing programs.

Choosing the second option, we decided that each sign record was to be thirteen 2-byte words, long enough to contain 24 fields (20 for notation and 4 for auxiliary information), and 2 bytes for a numerical index. Each record corresponds to the activity of one hand in a sign; thus a one-handed sign has one record, a two-handed sign two records, and compound signs as many records as their gestural components require.

A set of printable ASCII characters was selected to correspond to the DASL notation symbols, using graphic similarity or some other mnemonic for ease in coding. Because DASL does not arrange all transcription codes for a sign serially from left to right (but subscripts and stacks MOV symbols vertically), as needed for our purposes, we devised a coding grid. This grid permitted linearization and the linear transfer of codes to fields 2 through 22 of a record. It also made it possible to employ persons unfamiliar with DASL notation to perform this task.

Of the four fields reserved for auxiliary information, the first contains codes designating major sign types: one-handed, two-handed, compound, or non-compound. A second auxiliary field will contain codes specifying the type of variation of a sign: formal, informal, regional, etc., and various inflected forms. The remaining two fields are left blank at present. Figure 1 shows the grid pattern for fields to be coded, and Table 1 shows how information in a sign is arranged in 24 fields.

Figure 1 shows, for example, how the sign RESIST was coded on the grid and entered on the 16th record of the DASL file. The box numbered 10 relates to the interaction between the two hands. It appears four times on the coding grid because diacritics appear four places in DASL notation: above or below a HC, or as a subscript or a superscript to the HC. A problem arose here for two signs; they carried two interaction (box 10) characters): SHIT, _(35v, and MAGAZINE, BIC . It was resolved by placing the two interaction symbols in the same field in each of two records, since both signs are two-handed and require two records. When only one interaction symbol was given for a sign, it went to box 10 of the grid and record designating the non-dominant hand. Other similar problems were solved for signs that carry two movement modifiers, alternation and repetition, as in the sign BRAG (DASL p. 183), by placing them in field 20 on each of the two records; usually MOV symbols, including modifiers, are entered in the record for the dominant hand only.

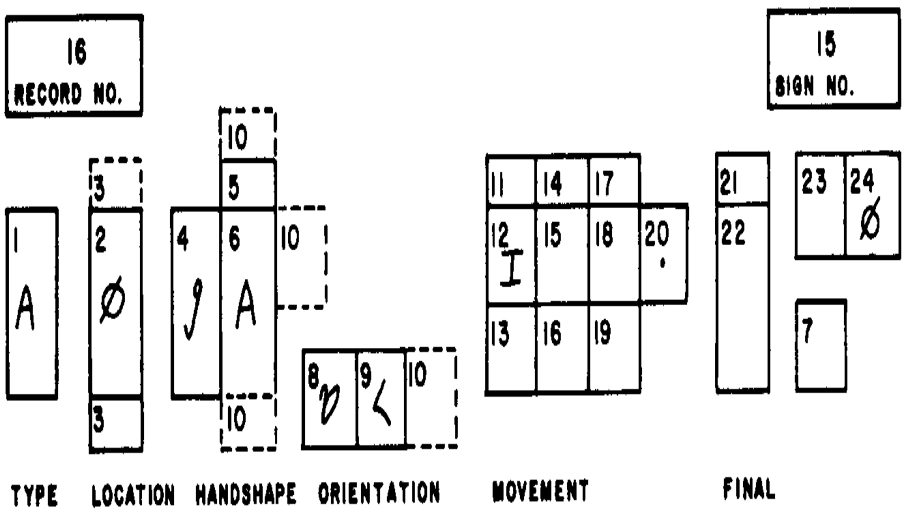



Figure 1 . Grid pattern for coding from DASL notation. The example given is the sign RESIST (DASL p. 3)

Note that the order of the hand orientation subscripts in the grid differs from the order in the notation. See Appendix C for the computer printout of the record of this sign.

| Field | Use for | Permitted characters |
|---|---|---|
| 1 | Sign type | A B C D E F G H I; 2 3 4 5 6 |
| 2 | Location | 0 O ^ = U 3 P [ \ J A D ; or blank if hand ; is location |
| 3 | Location diacritic | _ or blank |
| 4 | Forearm prominence | J or blank |
| 5 | Handshape modifier or initialized sign | . " ] # ; I (if initialized sign) or blank |
| 6 | Handshape | A B 5 C E F G H I K L 3 O R V W X Y 8 |
| 7 | | Blank for future use |
| 8 | Palm orientation | A D < > L T or blank |
| 9 | Finger direction | ^ V < > L T or blank |
| 10 | 2-Hand interaction | _ - 1 Q + 8 6 or blank |
| 11 | 1st mov't diacritic | . or blank |
| 12 | First movement | ^ V N > < Z T L I A D W 7 ] # % @ ) X 8 + 6 : " |
| 13 | 1st simult.movement | Characters as in Field 12 or blank |
| 14 | 2nd mov't diacritic | . or blank |
| 15 | Second movement | Characters as in Field 12 or blank |
| 16 | 2nd simult.movement | Characters as in Field 12 or blank |
| 17 | 3rd mov't diacritic | . or blank |
| 18 | Third movement | Characters as in Field 12 or blank |
| 19 | 3rd simult.movement | Characters as in Field 12 or blank |
| 20 | Movement modifier | . R ? or blank |
| 21 | Final HC diacritic | Characters as in Field 5 or 10 or blank |
| 22 | Final Handshape | Characters as in Field 6 or blank |
| 23 | | Blank for future use |
| 24 | Variant code | Numerical codes to be determined |

Table 1. Values of the 24 fields, with the ASCII codes allowed in the respective fields (Corresponding ASCII codes are shown below DASL characters if these differ).

The HC subscripts a and clearly define hand or palm orientation (as MOV symbols they denote supinating and pronating action respectively), and the subscripts V and Ashow the direction salient fingers are pointing. These subscripts are coded in fields 8 and 9 respectively. All other subscripts ambiguously designate either orientation, and were transferred as given in DASL (usually in field 8), when only one was indicated. Work is in progress on renotating each sign for PO and MO instead of direction of fingers.

For signs requiring multiple records, the two-handed and compound signs, we repeated on each record the information that DASL gives only once for the location or the first component of compound signs. When the location of a sign is the inactive hand, a blank character was typed in the location field, and the values of the inactive hand and hand orientations were assigned to the respective fields of the record for that hand .

**Hardware & software.**

The hardware used for the project includes a Digital Equipment Corporation PDP 11/10 with 16K words of memory, an RK-05 diskdrive, an RT-ll single-job monitor (Version 2C), and an LA-36 terminal console.

To begin our work we entered each sign that headed a lexical entry in DASL, unless it was identified as a variant (e.g. p. 5). The variants that reappeared as heads of separate entries in DASL were entered in our program as well. Our file now contains 1,768 of these "primary entries . " In the next step, we will determine the nature of each variation and code it as articulatory, dialectal, morphological, etc. Then these variants and other signs not listed as entries by themselves in DASL will be added to our file.

The information in each Dictionary entry was listed in three separate files. One file contains the transcription codes of the signs. These were entered and corrected from the console keyboard, using a specially written FORTRAN program EDITD; and since all records are of uniform length, the program uses a direct access file for both input and output and thus has quick access to individual records. See Appendices A and B for this program.

A second file contains word category information and glosses--a unique one-word (or hyphenated word) English translation--to aid the identification of transcribed signs on the printout.

A third file includes fuller English translations. More than one English translation may be given to each entry to cover its meanings. Sign numbers are used to reference the items in the three files. The contents of the second and third files were entered with a regular text editing program. Since the records in both these files are of variable length, the usual access to a record is sequential and requires more time than for a direct access file. But we created an index file, which gives us direct access to a gloss or a translation through an index to the beginning of the line containing it.

We envision a package of interactive programs, which users with little programming knowledge can employ to retrieve information from the files. For now, however, we have separate FORTRAN programs to perform three basic tasks:

o Serial listing of signs with both record and sign numbers, with or without glosses or translations (Appendix C)

o Detection of clerical errors by searching all fields for illegal characters

o Frequency count for signs with specified phonological characteristics

To count frequencies in DASL, the simplest search program reads in an array of 26 bytes (24 containing the transcription of a sign and 2 for the sign number) from the first record, and compares the characters in a specified field with a search character. If they are the same, it increments a counter and proceeds to the next record. Appendix D lists a program for searching and counting major sign types and its run, showing the result. When there are structural conditions to consider--for example, symmetry of signs--a series of carefully written FORTRAN if-statements is needed to sort out symmetric from asymmetric signs. Nearly all search programs we have used require one or more branchings.

Search programs sometimes turn up items with erroneous code or anomalies that do not quite fit the basic sign patterns. These include signs in which the two hands perform at two different locations (e.g. SICK, one hand at head, the other at chest; DREARY-JOB, one hand grasping nose, the other rotating in front of chest), and where the MOV is non-manual (blowing along the palm for UTTERLY-NOTHING; see DASL p. 149).

In addition to using programs to detect clerical errors, we also proofread printouts of the files. Beyond specially designed search programs, we have written a general purpose retrieval program to obtain lists of signs similar to an input model. This program can be used to retrieve minimal pairs: signs t hat differ in only one parameter from a model sign (see

Appendix E).

**Results.**

The search programs we have run to date have verified linguistic claims about our corpus, have listed signs that met certain criteria, and have found distributional patterns of formational values in various categories of signs. Some searches described below will illustrate our methods . First, we determined frequencies of the sign types. How many one-handed and two-handed signs and how many compounds are listed as primary entries in DASL? The run of program SRCHl in Appendix D shows 1,001 two-handed (type A) and 6Z7 one-handed non-compound signs (type B). Of the two-part compounds, there are 45 two-handed to two-handed (type C), 27 one-handed to one-handed (type D), 59 one-handed to two-handed (typeF), and 5 two- to one-handed (E). (G, H, and J) are three-part compounds. Signs occurring at or near the supine (a) or prone (~) wrist or hand were coded as one-handed, because each of them required only one record to store. There are 4 and 38 of them, respectively; however, they may behave more like two-handed signs, with one hand serving as the location for the dominant hand. If we group these with other two-handed signs, the frequency ratio of one-handed to two-handed signs is nearly one to two (627 minus 42 vs. 1,001 plus 42). Thus, two hands yield a vocabulary only twice as large as that of one hand alone. Of the two-handed non-compound signs, nearly half have one hand stationary . Our next searches tested the findings of other researchers, who made hand tallies of lexical items or made other less systematic observations. Bellugi and Klima in their count of more than 2,000 ASL signs found that:

only 35% involve the use of both hands as active. About 40% of the signs are made with one hand only, and another 25% are made with one hand acting on the other hand which remains stationary as a base. Thus, for almost two-thirds of these signs, one hand is used as the dominant hand.(l975:232)

As seen in Table 2, our computerized tally of 1,628 noncompound primary entries corresponds to the earlier estimates: 32% are symmetric in all respects but hand orientations; 1% are made by different handshapes, both active; 31% are made with one hand acting on the other hand passive; 36% are made with one hand only. Thus, 67% are made with one hand dominant. So far, we have a reliability check of two different methods of tallying frequencies.

| type | one hand active | | both hands active | | total | |
|---|---|---|---|---|---|---|
| One-handed signs | 585 | 36% | | | 585 | 36% |
| Two-handed signs: | | | | | | |
| Same HC (both) | 187 | 11%* | 525 | 32%* | 712 | 44% |
| Different HC | 315 | 19%* | 16 | 1%* | 331 | 20% |
| Sum | 502 | 31% | 541 | 33% | 1,043 | 64% |
| All Non-compound signs | | | | | 1,628 | 100% |

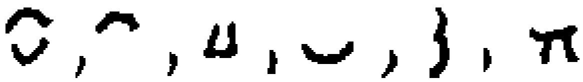
* $Chi^2 = 426.94$, $df = 1$, $p < .001$

Table 2. Frequency distribution of major structural types of signs, and percentages of 1,628 non-compound primary entries.

Table 2 shows in addition that among two-handed signs, there are twice as many signs where both HCs are the same as there are where they are different (44% vs. 20%). The number of signs with both hands active roughly equals the number of signs with only one hand active (33% vs. 31%). The frequencies match even more closely if we consider two-handed signs with HCs different and moving (N=16) to belong to the set of HC different and one hand active (N=315). In this set of 16 signs, the dominant hand seemingly pushes the non-dominant hand, thus effecting the motion of the latter. Examples of such signs are HELP, SHOW-IT, SUPPRESS, IMPRESS, MEAT, etc. We can generalize then and say that whenever both hands have different configurations, one hand is dominant over the other. Of the two-handed signs with the same HCs, about three-fourths have both hands moving, and one-fourth have one hand acting on the other. This inequality is explained by the restriction of the inactive hand to a smaller set of HCs than if both hands were to move symmetrically. A Chi-square test of the interaction between handshape match (rows) and hand dominance (columns) is highly significant ($X2=426.9$, $p <0.001$, $df =1$).

Another search, inspired by Siple 1978, asked if signs made farther away from the observer's hypothesized fixation point on the signer's lower face will have less marked (more neutral, simpler) handshapes and will be two-handed more often than those closer to it. Battison (1978:42) also found significant interaction between the simplicity of handshapes and the type of location, based on his count of 606 relevant signs.

To test this we established two categories based on markedness of handshapes: unmarked are those one-handed signs with the HC s B, G, A, 5, C, or O, or two-handed HCs; all others are marked. Battison (1978:36-39) offers several arguments for the unmarked nature of HCs B, G, A, 5, C, and O. The head and neck locations



classify the high acuity area, and all others the low acuity areas .

| | Head and Neck Locations | | | Off Head and Neck Locations | | |
|---|---|---|---|---|---|---|
| HC | 1-handed | 2-handed | Total | 1-handed | 2-handed | Total |
| Unmarked | 222[2] | 44[3] | 266[1] | 137[2] | 531[3] | 668[1] |
| Marked | 133[2] | 19[3] | 152[1] | 9[2] | 449[3] | 542[1] |
| Total | 35[4] | 63[4] | | 230[4] | 90[4] | |

[1] Chi-square = 8.685, $p < 0.005$, df = 1
[2] " = 0.40, no significant interaction
[3] " = 5.251, $p < 0.025$, df = 1
[4] " = 583.56, $p < 0.001$, df = 1

Table 3. Frequencies of signs, showing interaction between complexity of handshape, number of hands, and distance from observer's eye fixation point.

Table 3 shows a significant interaction between location and markedness (first footnote in table), much the same as reported by Battison, although he excluded from his count the signs made in "neutral" space and on the arm and hand. This interaction, however, does not hold for one-handed signs but only for two-handed signs (X2 = o . 40, and X2 = 5 . 25, p <

O . 025), respectively. In addition, there is a very significant interaction between number of hands and location (X~ = 5 . 83 . 5 6, df = 1 ) . Thus, there is a tendency for signs to be two-handed when the signing is in an area of low visual acuity, and one-handed when in high acuity areas. In fact, many formerly two-handed signs with identical HCs in high acuity areas have become one-handed; while in low acuity areas they remain two-handed; e.g. COW and CAT (2 hands) versus FINISH and HAVE (1 hand).

Mark Mandel has suggested to us another structural regularity about the complexity inherent in signs that change HCs during their execution (personal communication). The hypothesis that we tested was that HC change will tend not to co-occur with a bi-directional movement, but only with a uni-directional movement.

Table 4 shows that 68% of 187 HC-changing signs require some uni-directional movements, whereas virtually no sign permits any bi-directional motion with either opening or closing of the hands. Fifty-seven of 82 (70%) signs with opening MOV, and 71 of 105 (68%) with closing MOVs are articulated simultaneously with a uni-directional MOV, most frequently with forward MOV, followed by up and down MOVs. Other unidirectional MOVs are used less commonly in conjunction with opening (fewer than 5) and with closing (between 5 and 10 for other straight MOVs). Table 4 indicates that no other sign in DASL involves both closing and either supination or pronation. There is indeed a sign GRAB-UP (Ø $5_{ɒ}$ $5_{ɒ}{}^{\#}_{a}$), which requires simultaneous closing and supination. Thirteen signs with opening and 20 signs with closing MOV are listed as having just these movements (Alone in Tab. 4), and 12 signs with opening and 14 signs with closing require some other MOVs like touching, circling, and some kind of 2-hand interaction. These sets of signs may involve some uni-directional movements that are not explicitly shown in DASL.

| MOV | Opening | Closing | % of 187 |
|---|---|---|---|
| Alone | 13 | 20 | 18 |
| uni-directional MOVs | | | |
| Up | 16 | 11 | 14 |
| Down | 8 | 18 | 14 |
| Forward | 23 | 21 | 24 |
| Toward | 3 | 8 | 6 |
| Right | 2 | 7 | 5 |
| Left | 2 | 6 | 4 |
| Supinate | 1 | 0 | 1 |
| Pronate | 2 | 0 | 1 |
| uni-directional total | 57 | 71 | 68 |
| any bi-directional MOVs | 0 | 0 | 0 |
| column totals | 82 | 105 | 100 |

Table 4. Frequencies and percentages of signs having movements articulated simultaneously with opening and closing action.

Not listed in Table 4 are the results of a further search, the interaction of wiggling MOV with other MOVs. Bi-directional movements seem also to be incompatible with wiggling. Only the sign PIANO is listed with a bi-directional movement (right and left Ƶ )and wiggling.

We conclude that a change in HC as sign action usually involves another motion, more often of a linear nature, but it will exclude bi-directional MOVs and possibly certain 2-hand interactional MOVs; e.g. entering while opening. At present, we suggest in more general terms that ease of articulation for HC changes requires an accompanying movement and limits the options to a smaller set.

**Development.**

Much could be added to our dictionary: a more detailed transcription of articulation, variations in pronunciation, lexical non-manual aspects such as facial expressions, morphological modifications, finer grammatical classifications (beyond the DASL categorization into noun, verb, and other), etymologies, synonyms, semantic properties, and usage. Sub-phonemic questions concerning, e.g., the complementary distribution of "allophones,' cannot be addressed in our computerized DASL without amending the transcription system. However, one can obtain from the computer a list of signs that meet some criteria in terms of DASL notation, and then sift the list to obtain the subset. This is what Mark Mandel did for his kinesiological studies on knuckle-wrist connections in signs after we supplied him with some lists (Mandel 1979).

The fact that sometimes it is necessary to screen out signs that do not meet a criterion, although they bear the same transcription used for the criterion as do other signs, points to the need for renotating some signs without leaving the DASL system. For example, DASL lists the signs NOTHING Ø✓F✓F ≷ and PERFORM Ø✓$C_D$✓$C_D$ ≷ with the same MOV, but the movement is different in the two signs: the former is always enacted symmetrically [both upper arms rotate medially then laterally], the latter often with parallel movement one arm rotating medially while the other rotates laterally, then the opposite]. DASL, however, lists a symmetric variant of PERFORM, noted more appropriately with the MOV )( Another instance of ambiguity in MOV notations is the one for interchange, . This symbol is used to indicate either full MOV or movement modifier; as a MOV it can either include the rotation of both hands, more like the signs with MOV or it may not imply such rotation but only reversal of the position of the hands (see DASL, p. 60 for signs SUBSTITUTE; and INTERPRET). As a modifier, this MOV symbol specifies a type of alternation, i.e. the alternating interaction of one hand with the other (see DASL, p. 36 for signs BUILD, PEACE, and HANDS). Compound signs also need a closer look. Many of them were notated as if the components were articulated fully, and as if no deletion or simplification of movement occurred. [For a thorough treatment of compound signs in ASL, see Klima & Bellugi 1979, ch. 9. Ed. note.]

This leads us to what we are going to do next. It may be helpful to outline what our plans are. We will:

o renotate some signs to improve the consistency of the use of DASL notation and renotate compounds to include the deletion of movement in compounding

o in a separate sign file specify fully the allophones used, hand orientation, points of contact, and final handshapes

o enter variations and modifications that exist for each lexical item; for this we need to classify usage, status of pronunciation variations, morphology and syntax, and attach codes to each sign to indicate its membership in the various categories

o collect signs not noted in DASL together with all known information on each of them.

We are also interested in making this resource available to sign language researchers who have an inquiry that calls for a search of the DASL files. We welcome the readers' input to the development of the data-bank, concerning especially phonetic notation, grammatical classifications, sociolinguistic categories, and other lexical aspects. We would also like to receive for inclusion in our computer dictionary lexical items not found in DASL, be they single signs, compounds, idioms, or sentence examples, and additional meanings of signs listed in DASL but not given there.

**Notes.**

1. This study has been supported in part by National Science Foundation Grant 768-2530 and NIH Grant NS 14923. We gratefully acknowledge contributions by the following persons: Francois Grosjean, Steve Reef, Howard Poizner, Lolita Sabine, June Nunes, Marie Philip, Ella Lentz, Craig Andree, Maryann Sacco, Bill Flamburis, Mark Mandel, and William Stokoe.

2. While this was being prepared, a paper was received from David V. Moffat, University of North Carolina, in which he outlined his method of computerizing a sample of 153 signs from DASL, using APL language. He selected APL because its character set matches closely in graphic form with the DASL notation characters.

**References**

Battison, R.

1978 Lexical Borrowing in American Sign Language (Silver Spring, MD, Linstok Press).

Bellugi , U ., & E . Klima

1975 Aspects of Sign Language & Its Structure, in <u>The Role</u> <u>of Speech in Language,</u> Kavanagh & Cutting eds. (Cambridge, MA, MIT Press).

Grosjean, F., H. Teuber, & H. Lane

1979 When is a Sign a Sign: The On-Line Processing of Gated Signs in American Sign Language. Unpub. MS, Northeastern University, Boston.

Klima , E ., & U. Bellugi

1979 <u>The Signs of Language</u> (Cambridge, MA, Harvard U. Press).

Mandel, M .

1979 Natural Constraints in Sign Language "Phonology": Data from Anatomy, <u>Sign Language Studies</u> 24, 215-229.

Moffat, D.

1979 A Sign Language Dictionary on the Computer. Unpub. MS, University of North Carolina, Chapel Hill.

Poizner, H., & H. Lane

1978 Discrimination of Location in American Sign Language, in <u>Understanding Language through Sign</u> <u>Language Research,</u> Siple ed. (NY, Academic Press).

Poizner, H., R. Battison, & H. Lane

1980 Cerebral Asymmetry for American Sign Language: The Effect of Moving Stimuli, Brain & Language 7, 351-362.

Siple, P.

1978 Constraints for Sign Language from Visual Perception Data, Sign Language Studies 19, 97-112.

Stokoe, W.

1960 Sign Language Structure: An Outline of the Visual Communication Systems of the American Deaf (= Studies in Linguistics: Occasional Papers 8), rev. 1978 (Silver Spring, MD, Linstok Press).

Stokoe, W., D. Casterline, & C. Croneberg

1978 A Dictionary of American Sign Language on Linguistic Principles (Washington, Gallaudet College Press); revised 1978 (Silver Spring, MD, Linstok Press).

Stungis, J.

1980 Identification and Discrimination of Handshape in American Sign Language. Technical Report, Northeastern University.

APPENDIX A. Listing of Program "EIDITD"

```
      LOGICAL*1 IW(24),IOP,COMMAND(5)
      DATA COMMAND/'E','R','D','A','C'/
      CALL ASSIGN(1,'DK:DASL.DIC',-1)
      DEFINE FILE 1(10240,13,U,IR)
      TYPE 40
40    FORMAT(' COMMANDS ARE: A = ADD, C = CHANGE'/
     1' D = DELETE, E = EXIT, R = RETRY')
      TYPE 41
41    FORMAT(' A ZERO LINE NUMBER STOPS PROGRAM')
2     TYPE 11
11    FORMAT('$RECORD NO.? ')
      ACCEPT 22,IR
22    FORMAT(I5)
      IF(IR.LE.0) STOP ' PROGRAM TERMINATED'
      IF(IR.LE.10240) GOTO25
      TYPE 6
6     FORMAT('$RECORD NO. TOO BIG, MAX IS 10240 ')
      GOTO 2
25    READ(1'IR,ERR=90,END=90)(IW(I),I=1,24),IS
      IR=IR-1
      TYPE 8,(IW(I),I=1,24),IS
8     FORMAT(1X,24A1,I5)
20    TYPE 9
9     FORMAT('$WHAT COMMAND? ')
      ACCEPT 10,IOP
10    FORMAT(A1)
      DO 7 J=1,5
      IF(IOP.EQ.COMMAND(J)) GOTO (1,2,3,4,5),J
7     CONTINUE
      TYPE 19
19    FORMAT(' BAD COMMAND, TRY AGAIN')
      GOTO 20
3     DO 13 I=1,24
13    IW(I)="40
      IS=0
17    WRITE(1'IR,ERR=90,END=90)(IW(I),I=1,24),IS
      GOTO 2
4     TYPE 11
      ACCEPT 22,IR
5     TYPE 15
15    FORMAT(' 123456789012345678901234 INDEX')
      ACCEPT 16,(IW(I),I=1,24),IS
16    FORMAT(24A1,X,I5)
      GOTO 17
90    ENDFILE 1
1     STOP ' PROGRAM TERMINATED'
      END
```

# APPENDIX B. Run of Program "EDITD"

```
Underlined portions typed by user.

.R EDITD<CR>
*DASL.DIC<CR>

COMMANDS ARE: A = ADD, C = CHANGE
D = DELETE, E = EXIT, R = RETRY      Explanation:
A ZERO LINE NUMBER STOPS PROGRAM

RECORD NO.?    16<CR>                Enter sign #13 on record #16

WHAT COMMAND? A<CR>                  Add command

RECORD NO.?    16<CR>

12345678 9012345678901234 INDEX
A0 J D<    I       .       15<CR>    Enter codes incorrectly

RECORD NO.?    16<CR>                Redo

A0 J D<    I       .          15

WHAT COMMAND? C<CR>                  Change command

123456789012345678901234 INDEX
A\A\B0 J A D<  I       .      15<CR> Retyped correctly incl.rubouts

RECORD NO.?    16<CR>                Check if correct

B0 J A D<  I       .         15      Proofread, correct!

WHAT COMMAND? R<CR>                  Move on to next entry

RECORD NO.?    10<CR>                Error in record #10

B0   A    TV                 9       sig "TV" in wrong fields
WHAT COMMAND? C<CR>

123456789012345678901234 INDEX
B0   A     TV            9<CR>       Correction
RECORD NO.?    25<CR>

B0 J A     7                 24
WHAT COMMAND? D<CR>                  Delete, record #25 now blank

RECORD NO.?    26<CR>

B0   A L   ]       R         25
WHAT COMMAND? B<CR>                  Unrecognized command

BAD COMMAND, TRY AGAIN

WHAT COMMAND? E<CR>                  End of session
STOP -- PROGRAM TERMINATED
```

APPENDIX C. Run of Program "SINGLO" for listing both signs and English glosses.

```
.R SINGLO<CR>
*DASL.DIC<CR>                       Name of sign file
*GLOSS.DIC<CR>                      Name of gloss file

1,26<CR>                            Beginning and ending record numbers

Rec.# |----Sign-Notation----| Sign # | Categ.| Gloss

      123456789012345678901234
    2 B0  .A     ^                  2 0  NVX   CHIEF-PERSON
    3 D0  .A     ^                  3 0  N     CHIEF-IN-AREA
    4 20   B D   @                  3 0
    5 B0  .A     ^                  4 0     S  -ER (COMPARATIVE)
    6 B0   A     V                  5 0  N X   SOUTH
    7 B0   A     V                  6 0    X   TUESDAYS
    8 B0 J A   V N                  7 0  NV    SUITCASE
    9 B0 J A    .T                  8 0   V    REFUSE
   10 B0   A     TV                 9 0  NV    GEARSHIFT
   11 B0  .A     L                 10 0  N     SELF
   12 B0  .A    .L                 11 0        SO-LONG
   13 B0  .A     L       .         12 0  NV    DOORBELL
   14 B0   A L   L       .         13 0  NV    KNOCK
   15 B0  .A     I       .         14 0  NV    BACK-AND-FORTH
   16 B0 J A D<  I       .         15 0  NV    RESIST
   17 B0   A     A                 16 0     S  POSSESSIVE-'S
   18 B0  .A    .A                 17 0  N X   TEN
   19 B0  .A     AV                18 0  N X   ANY
   20 B0  .A     A                 19 0  N X   ANOTHER
   21 B0 J A T   D>                20 0  NV    REBEL
   22 B0 J A     A<                21 0   V    POUR
   23 B0 J A     W                 22 0   V    HEADSHAKE-NO
   24 B0   A L   7       .         23 0    X   YES
   25 B0 J A     7                 24 0   V    HEAD-BENDING-DOWN
   26 B0   A L   ]       R         25 0  NV    FLASH-LETTERS

^C
```

APPENDIX D. Listing of Program "SRCH 1" to find the frequency of sign-types as coded in field one.

```
C       SEARCHES AND COUNTS CHARACTERS IN COL 1
        LOGICAL*1 IW(24),ITYPE(9),IB
        DATA IB,ITYPE/' ','B','D','A','C','F','E','H','J','G'/
        DIMENSION ICOUNT(9)
        NN=0
        DO1 I=1,9
 1      ICOUNT(I)=0
        CALL ASSIGN(1,'DASL.DIC',-10,'RDO')
        DEFINE FILE1(3092,13,U,IR)
        IR=1
        DO2 I=1,3092
        READ(1'IR,END=15)(IW(K),K=1,24),IS
        DO3 J=1,9
        IF(IW(1).EQ.ITYPE(J))ICOUNT(J)=ICOUNT(J)+1
 3      CONTINUE
        IF(IW(1).EQ.ITYPE(3).AND.IW(2).EQ.IB)NN=NN+1
 2      CONTINUE
 15     TYPE5,NN
        TYPE4,((ITYPE(I),ICOUNT(I)),I=1,9)
 5      FORMAT(' RESULT OF SEARCH AND COUNT OF SIGN-TYPES AS GIVEN
       1IN COLUMN ONE',//,' NUMBER OF SIGNS WITH DEZ AS TAB:',I4,/)
 4      FORMAT(' NUMBER OF SIGNS OF TYPE ',A1,2H: ,I5)
        STOP
        END
```

Run of "SRCH1". Underlined portion typed by user.

```
.R SRCH1<CR>
*DASL>DIC<CR>

RESULT OF SEARCH AND COUNT OF SIGN-TYPES AS GIVEN IN COLUMN ONE

NUMBER OF SIGNS WITH DEZ AS TAB:  456

NUMBER OF SIGNS OF TYPE B:   627
NUMBER OF SIGNS OF TYPE D:    27
NUMBER OF SIGNS OF TYPE A:  1001
NUMBER OF SIGNS OF TYPE C:    45
NUMBER OF SIGNS OF TYPE F:    59
NUMBER OF SIGNS OF TYPE E:     5
NUMBER OF SIGNS OF TYPE H:     2
NUMBER OF SIGNS OF TYPE J:     1
NUMBER OF SIGNS OF TYPE G:     1
STOP --
```

APPENDIX E. Run of general purpose retrieval program "RETRIV."
This shows retrieval of signs prefixed (in DASL) with , and signs with opening and another simultaneous movement as a second MOV.

```
.R RETRIV<CR>
*DASL.DIC<CR>

ENTER YOUR MODEL SIGN UP TO 4 RECORDS.* MEANS "ANY".ENDS WITH "!"
1234567890123456789012
*^   G **  X          !<CR>

S#  744  B^   G T   X              NV  THINK
S#  745  F^   G T   X               V  FAINT
S#  746  F^   G T   X               V  HOPE
S#  748  F^   G T   X              NV  FAITH
S#  749  F^   G T   X              NV  BELIEF
S#  750  F^   G T   X              NVX DECIDE
S#  751  F^   G T   X              NV  AGREE
S#  752  F^   G T   X              NV  DISAGREE
S#  753  F^   G T   X               V  TWIST-TRUTH
S#  754  F^   G T   X                X NARROW-MINDED
S#  755  F^   G T   X                X BROAD-MINDED
S#  756  F^   G T   X              N X ABSENT-MINDED
S#  757  F^   G T   X              NV  GOAL
S#  758  F^   G T   X                X GULLIBLE
 15 SIGNS RETRIEVED

ENTER YOUR MODEL SIGN UP TO 4 RECORDS.* MEANS "ANY".ENDS WITH "!"
1234567890123456789012
****** *******]******!<CR>

S#   87  20   A      ) .]X         NV  CLASH (VAR. OF 538)
S#   90  20   A      X  ]:           X WITHOUT
S#  262  20   5 A    # .]     .      X STICKY
S#  364  20   F      X  ]L           X USELESS
S#  538  20   O <    ) .]X    B    NV  CLASH
S# 1004  BU  #V T    X  ]^         N   GOAT
S# 1346  2    A      VX ]V          V  REMOVE
S# 1576  2    O A    X  ]^    .      X RICH
S# 1579  2    O A    X  ]L         NVX WASTE
S# 1580  2    O A    X .]D           X EXPENSIVE
  10 SIGNS RETRIEVED

ENTER YOUR MODEL SIGN UP TO 4 RECORDS.* MEANS "ANY".ENDS WITH !
1234567890123456789012
^C
```

Hartmut Teuber earned his degree in Education of the Deaf at Gallaudet College and taught for several years. He is now responsible for maintenance and expansion of the computerized dictionary project, is working on phonetic and orthographic transcription systems for signs, also designs sign stimuli for psycholinguistic experiments.

Robbin Battison worked with American Sign Language at the Salk Institute for Biological Studies, Linguistic Research Laboratory at Gallaudet College, and Language Perception Laboratory at Northeastern University. He is now the Manager of the Document Design Center at the American Institutes for Research in Washington, D.C.

Harlan Lane is a co-principal investigator of an NIH grant to study the perception and production of ASL. His interests go beyond comparative psycholinguistic studies of speech and sign as he is preparing a book on the social history of the deaf and their language. His earlier book, The Wild Boy of Aveyron, dealt with the early history of sign language and the genesis of special education.

Joe Heck, who played a major role in the hardware and software implementation of the project as departmental computer technician, earned his degree in Electrical Engineering at Northeastern University. Now he is a Senior System Programmer at the central computing center at Northeastern.

Jim Stungis, who designed the original 24-field coding grid and structured the matching codes for the DASL symbols, his nearly completed his doctoral degree in Psychology at Northeastern University. He is working on developing distinctive feature models for handshapes and movements of American Sign Language, as well as on spatial frequency studies of signs.

**THE ETYMOLOGY OF AN ESOTERIC SIGN**